PRELIMINARY ROUND RTL DESIGN TEST - FPGA EXTENDED CONTEST 2026

Watchdog Monitor (TPS3431-like) + UART Configuration

Platform: Kiwi 1P5 Board (Gowin GW1N-UV1P5)

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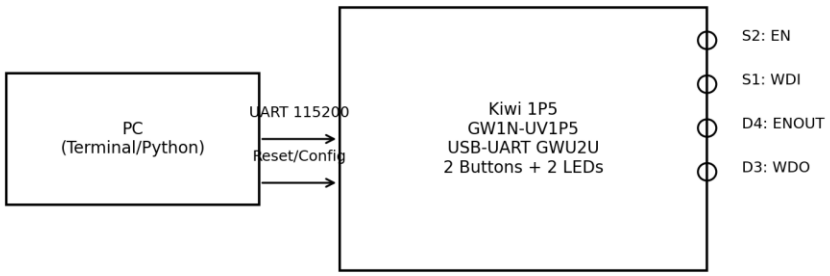
Minh hoạ setup test trên board



Note: This assignment requires emulating the watchdog principle of the TPS3431 IC, while allowing runtime configuration of operating parameters via UART on the Kiwi 1P5 board.

1. Objective

Design an RTL block (Verilog/SystemVerilog) that implements a watchdog supervisor function similar to TPS3431:
- Monitor the kick signal (WDI) and detect timeouts.
- Output a fault/reset signal (WDO) as active-low (open-drain emulation).
- Support watchdog Enable/Disable and the ENOUT signal.
- Allow runtime configuration of parameters (tWD, tRST, arm_delay...) via UART.

2. References

1) TPS3431 Watchdog Timer Datasheet (Texas Instruments, Rev Oct 2021).

2) Kiwi 1P5 FPGA Board - Quick Start Guide (OneKiwi/Gowin).

Kiwi 1P5 board link: https://www.gowinsemi.com/upload/database_doc/3224/document/68d5f9f48daf0.pdf

3. System Functional Description

The following functional requirements will be used as grading criteria:

- WDI is sampled on the falling edge. Each falling edge resets the watchdog counter.
- If there is no WDI falling edge within tWD, WDO must assert (pulled low) for tRST, then release and start a new cycle.
- EN=0: watchdog disabled, WDI ignored; ENOUT is 0; WDO is released (hi-Z/pull-up).
- When the watchdog transitions from disabled to enabled, WDI is ignored during the arm_delay window (default 150 us).
- Reference (typical) values: tRST ~ 200 ms; tWD preset ~ 1600 ms (CWD=NC) or ~ 200 ms (CWD=10 kΩ pulled up to VDD).

3.1 Default parameters after reset (recommended)

Parameter	Default value	Notes
tWD_ms	1600	Emulates CWD=NC mode (typ. 1600 ms)
tRST_ms	200	WDO hold time during fault (typ. 200 ms)
arm_delay_us	150	WDI ignore window right after enabling the watchdog

4. Detailed Functional Requirements

4.1 Enable/Disable and ENOUT

The design must satisfy:

1. Reset behavior: after system reset, the watchdog is in the disabled (safe) state.
2. EN=0: watchdog disabled, WDI ignored; ENOUT=0; WDO released (no fault).
3. EN transitions 0→1: start counting arm_delay_us; during this window all WDI kicks are ignored.
4. After arm_delay_us: the watchdog starts counting tWD; ENOUT=1 (indicates the system is allowed to run).

4.2 Watchdog Kick and Timeout

The design must satisfy:

5. A valid kick is a 'falling edge' event of WDI (from button S1 or UART KICK).
6. Each valid kick resets the tWD timer to 0.
7. If tWD expires before any kick: assert WDO=0 for tRST_ms; after tRST_ms, release WDO and start a new cycle.
8. Optionally allow writing CLR_FAULT to release WDO immediately (if implemented, describe clearly in the README).

4.3 Open-Drain Emulation

TPS3431 uses open-drain outputs for WDO and ENOUT. On an FPGA, this can be emulated in one of two ways:
A) Output a 'drive_low' signal and use tri-state (inout) if the tool flow/board supports it.
B) Simplify: use push-pull outputs, but keep the active-low convention for WDO and valid logic for ENOUT.
Contestants must clearly state the chosen approach in the README and ensure the LED demo works.

5. UART Parameter Configuration

Default UART: 9600 bps, 8N1. The design must support reading/writing configuration registers and the kick command via UART.

5.1 Frame Protocol (recommended for the contest)

Frame: [0x55][CMD][ADDR][LEN][DATA...][CHK]
- CHK = XOR of all bytes from CMD through the end of DATA.
- The design must return an ACK/RESP via UART (at minimum return STATUS or echo).

5.2 Mandatory Register Map

Addr	Name	R/W	Width	Description
0x00	CTRL	R/W	32	bit0 EN_SW; bit1 WDI_SRC; bit2 CLR_FAULT (write-1-to-clear)
0x04	tWD_ms	R/W	32	Watchdog timeout (ms)
0x08	tRST_ms	R/W	32	WDO hold time during fault (ms)
0x0C	arm_delay_us	R/W	16	Ignore WDI after enable (us)

0x10	STATUS	R	32	bit0 EN_EFFECTIVE; bit1 FAULT_ACTIVE; bit2 ENOUT; bit3 WDO; bit4 LAST_KICK_SRC

5.3 Minimum UART Commands

- WRITE_REG (0x01): write register.
- READ_REG (0x02): read register.
- KICK (0x03): generate one internal kick event (equivalent to one WDI falling edge).
- GET_STATUS (0x04): quick STATUS response.

6. Pin Mapping and Resources on Kiwi 1P5

Use the on-board resources for a visual demo (buttons/LEDs) and UART configuration.

Function	Board resource	Net/IO (per guide)	Notes
WDI (kick)	Button S1	IOR1B (KEY1)	Active-low; debounce minimum 5-20 ms
EN	Button S2	IOR1A (KEY2)	Active-low
WDO	LED D3	IOR17A (LED1)	Fault indication (convention chosen by contestant)
ENOUT	LED D4	IOR15B (LED2)	Turns on after enable + arm_delay
UART	USB-UART GWU2U	RXD/TXD (refer to Fig 02)	Used for PC configuration; contestant declares constraints per pinout mapping

Note: *Some special pins (JTAGSEL_N/RECONFIG_N and JTAG pins) should not be used as GPIO. See the pinout table in the Kiwi 1P5 documentation.*

7. Suggested RTL Architecture (reference, not mandatory)

Contestants may split the design into modules such as:

- uart_rx / uart_tx: UART RX/TX at 115200 8N1.
- uart_frame_parser: frame decode + checksum, update registers.
- regfile: stores CTRL/tWD/tRST/arm_delay and STATUS.
- sync_debounce: synchronizer + debounce for S1/S2.
- watchdog_core: FSM + timers; accepts kicks; drives WDO/ENOUT.

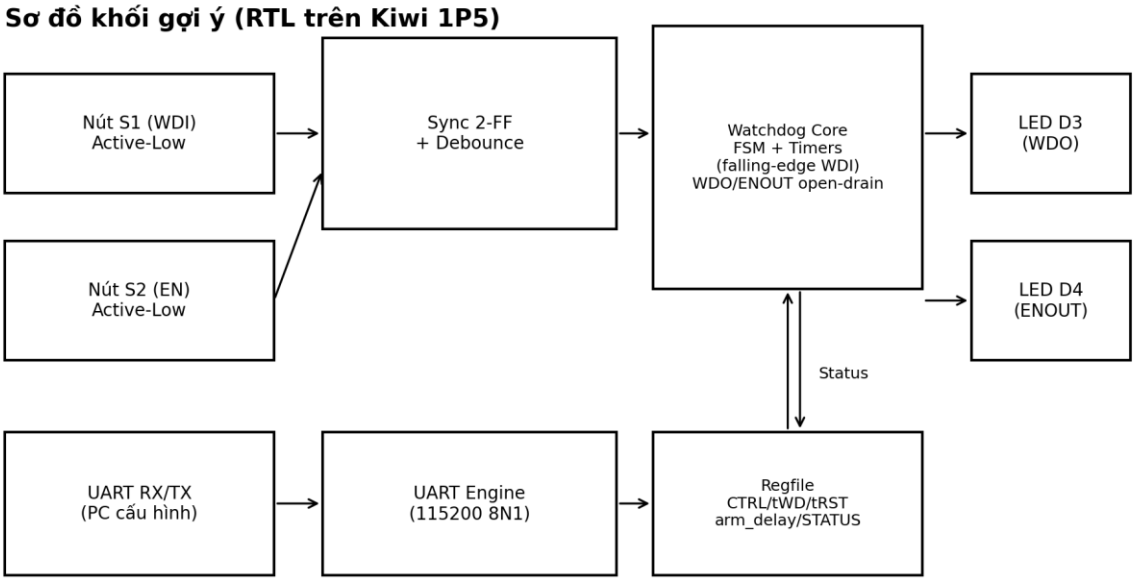


Figure 1: Suggested block diagram for the contest

8. Timing Diagram (reference)

The figure below illustrates a typical scenario: EN is enabled, ENOUT goes high after tRST/arm_delay, WDI is kicked a few times, then missing kicks lead to a timeout and WDO is pulled low for tRST.

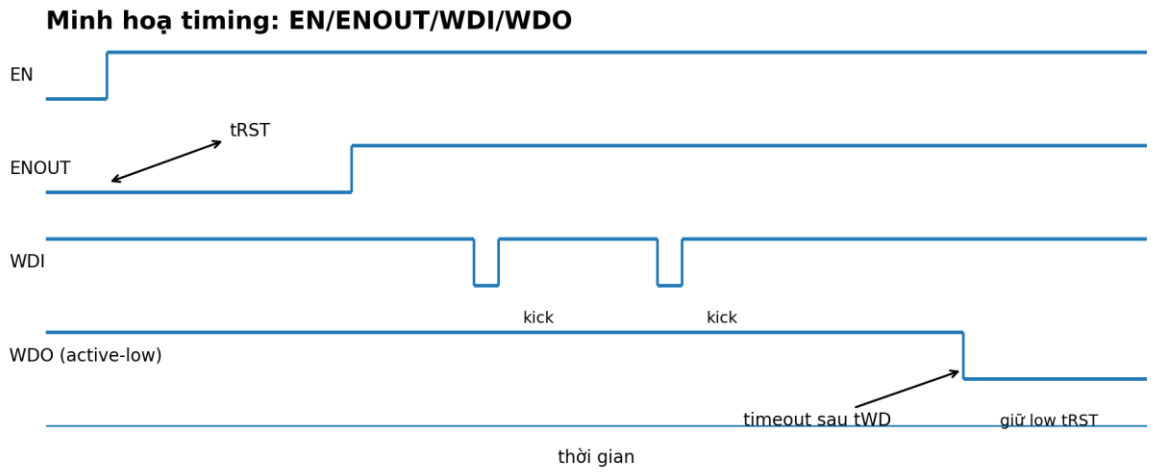


Figure 2: Timing diagram of EN/ENOUT/WDI/WDO

9. Deliverables

- Complete RTL (Verilog/SystemVerilog) + top-level that runs on Kiwi 1P5.
- Constraints file with correct pin assignments (buttons/LEDs/UART per pinout mapping).
- Testbench covering at least 5 cases: normal kick, timeout, disable, disable→enable (arm_delay), and parameter changes via UART.
- README describing how to build, how to run the demo on the board, and the LED display convention.

10. Scoring Criteria

Item	Weight	Notes
Watchdog functionality (FSM + timing)	50%	Meets Section 4 requirements
UART + register map	25%	Stable R/W, checksum/ACK
Testbench	15%	Covers cases and corner cases
Code & documentation quality	10%	Coding style, README, easy to build/run